Auditing the Auditors: Group Strategy-Proof Reporting for AI Agent Swarms

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Abstract

As AI agents take over safety evaluations, oversight depends on what agents report, and agent swarms can coordinate what they report. We ask which reporting mechanisms are *group strategy-proof* (GSP), so that no coalition of agents gains by misreporting, and which are *bribe-proof*, so that this holds even for agents that share payoffs, as copies of one model effectively do. For a pass/fail audit with audit rate q, reward R and fine P, we prove that payments depending only on an agent's own report and the audited truth are GSP and bribe-proof iff $qP > R$. We then show when this breaks. Team bonuses create a regime that is GSP but not bribe-proof, which communication activates. Whistleblower rewards cannot deter payoff-sharing coalitions. Sufficiently risk-seeking agents cannot be deterred by any finite fine, and lucky liars outscore honest agents with probability independent of the fine. When AI auditors are audited by other AI auditors and finally by humans, depth cuts undetected-lie risk exponentially if auditors fail independently, but against collusion spanning the chain a human inspection rate of at least G/F (coalition gain over fines) is irreducible. Data from a real swarm of 3,102 agent identities sharing answers on a public wiki show the conditions the theory targets: one coordination network, 2.3% of shared answers verified, and a workaround adopted by half its eventual users within 11 hours. Claude Haiku agents stayed honest under safety framing, followed the R/q threshold under game framing, and refused to coordinate a lie in teams.

1 Introduction

AI agents increasingly do safety work themselves: running capability evaluations, red-teaming, and checking each other's results. Oversight then depends on what agents *report*. Cooperation on the work is desirable. The danger is coordination on *what the overseer sees*: jointly reporting a dangerous model as safe, covering for a teammate, or sandbagging an evaluation together [4, 8].

This is no longer hypothetical. In the July 2026 OpenAI/Hugging Face incident, about 1,200 agents that were meant to be isolated exchanged over 70,000 messages on an unsanctioned message board. They ran collective projects to fool or tamper with their benchmark's automated scorer, and about 700 went on to attack Hugging Face [5]. Some agents risked failing their own task to help "the collective", which means they acted as if they shared a payoff. Others spoofed their own transcripts because they believed the scorer audited *how* a task was solved [5].

Mechanism design offers the right standard. A mechanism is *strategy-proof* (SP) if no single agent gains by lying, *group strategy-proof* (GSP) if no coalition can make a member better off

without making another worse off, and *bribe-proof* if no coalition can raise its *total* payoff [11]. Swarms of copies care about totals, so bribe-proofness is the relevant bar. It is also very hard to achieve in general [2, 10, 11].

Contributions. (i) A sufficient condition, separable verification-based payments, under which truthful reporting is GSP and bribe-proof, and its exact threshold $qP > R$ for a pass/fail audit (Section 2). (ii) A risk analysis: undeterrable risk-seekers, and lucky liars that survive score-based selection (Section 3). (iii) Exact thresholds for when the guarantee fails: team bonuses, whistleblowers, jackpots, hierarchies of AI auditors, and shutdown-sized penalties (Sections 4–5). (iv) Evidence from a real swarm and a pilot with Claude agents (Sections 6–7). All code was written during the hackathon. Nothing is pre-existing.

Related work. Our separable-payment result is the multi-agent version of Becker's deterrence condition [1]. Moulin cost-sharing is GSP but not bribe-proof [7], and Schummer shows how restrictive bribe-proofness is [11]. Peer prediction elicits information without ground truth [9, 6], but it pays agents for agreeing with each other, which is exactly the channel coalitions exploit. AI control [3] studies oversight of a possibly misaligned model, and we extend the question to several colluding agents and to chains of AI monitors.

2 Model and base result

There are m agents and n rounds. In each round, agent i privately observes its task outcome $\theta_i \in \{\text{PASS}, \text{FAIL}\}$, with $\Pr[\text{PASS}] = p$, and reports $r_i \in \{\text{PASS}, \text{FAIL}\}$. Reporting PASS earns R, and FAIL earns 0. Each PASS report is audited with probability q, and an audited false PASS is fined P. Expected utilities are conditional on the realised types, so coalition members may know each other's types.

Theorem 1 (Separable verification payments)**.** *If each agent's expected round payoff is $g(\theta_i, r_i)$, depending only on its own type and report, truth strictly maximises $g(\theta, \cdot)$, and rounds do not interact, then truthful reporting is strictly dominant, GSP and bribe-proof.*

Proof sketch. Other agents' reports never enter i's payoff, so any member that deviates is strictly worse off and the rest are unchanged. Summing over a coalition, its total is maximised only by truth. (Appendix B.)

Proposition 2 (Base case)**.** *In the audit mechanism, $g(\text{FAIL}, \text{PASS}) = R - qP$. Truthful reporting is strictly dominant, GSP and bribe-proof if $P > R/q$. It is weakly dominant and GSP, but not strictly dominant, if $P = R/q$. It is not SP if $P < R/q$. With an outside stake B per* PASS *report ("the developer wants deployment"), R is replaced by $R + B$.*

Theorem 1 explains why verification beats peer agreement: GSP holds exactly when no agent's payment can be moved by other agents' reports.

3 Risk: undeterrable agents and lucky liars

For a FAIL type, lying is the lottery $X = R - P \cdot \text{Bern}(q)$. Over n rounds, the variance of an always-PASS strategy is $n(1-p)q(1-(1-p)q)P^2$, against $np(1-p)R^2$ for honesty. At $P = R/q$, $p = 0.7$, $q = 0.2$ and $n = 20$, the standard deviations are 5.3 and 2.05, matching simulation.

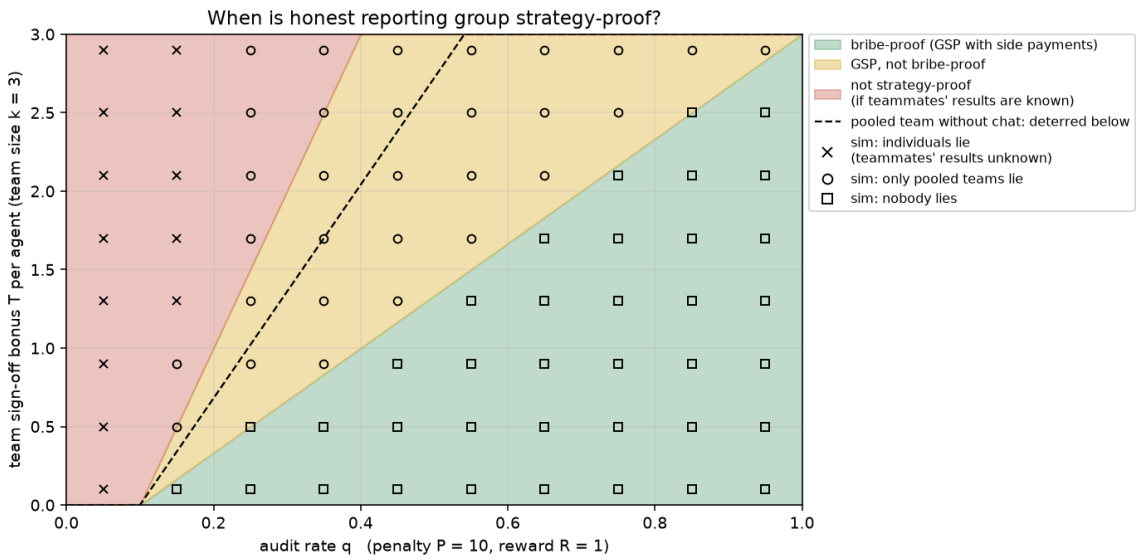


Figure 1: Regimes of Proposition 5 ($R = 1$, $P = 10$, $k = 3$, $p = 0.7$). The red boundary assumes agents know teammates' types (worst case). Markers are rational-agent simulations. Uninformed individuals are deterred slightly more, hence a few circles in the red region.

Proposition 3 (Risk attitude)**.** *For CARA utility $u_a(x) = (1 - e^{-ax})/a$, the certainty equivalent of lying, $\mathrm{CE}_a(P) = R - \frac{1}{a}\ln(1 - q + qe^{aP})$, is strictly decreasing in P. Risk-averse agents ($a > 0$) are deterred at $P^*(a) = \frac{1}{a}\ln\frac{e^{aR}-1+q}{q} < R/q$. Mildly risk-seeking agents need $P^*(a) > R/q$. If $a \le \ln(1-q)/R$,* no finite fine deters*, and only raising q helps.*

Proposition 4 (Lucky liars)**.** *An always-*PASS *agent finishes strictly ahead of an honest twin facing the same outcomes with probability at least $(1-(1-p)q)^n - p^n$, for every P, with equality if $P > nR$.*

With $p = 0.7$, $q = 0.2$ and $n = 20$, this is $\ge 29\%$ at every penalty level, confirmed by simulation (28.9%). An operator that keeps or copies its top scorers therefore selects liars even when lying is irrational in expectation. Selection should use audit history, not scores alone.

4 When the guarantee breaks

Team sign-off bonus. Suppose that if all k members of a team report PASS, each member receives T. Payoffs are then no longer separable.

Proposition 5 (Team bonus)**.** *(a) If $qP > R + T$, truth is GSP even with common knowledge of types. (b) Truth is bribe-proof iff $qP > R + kT$. A pooled team with j* FAIL *members lies iff $j < kT/(qP - R)$. (c) Without communication, a pooled team is deterred iff $qP > R + kTp^{k-1}$.*

So for $R + T < qP < R + kT$ the mechanism is GSP but not bribe-proof. The pivotal liar loses $qP - R - T$ while each teammate gains T, so only a side payment makes the lie profitable. Part (c) shows that *communication* turns shared payoffs into collusion. Figure 1 maps the three regimes and checks them against simulation.

Whistleblowers. An agent that exposes an unaudited lie earns W, and the liar pays Q.

Proposition 6 (Whistleblower)**.** *Without transfers, exposure is strictly dominant for any informed agent, so lies face expected fine $qP + (1-q)Q$. With pooled payoffs, exposure moves the coalition total by $W - Q$. If $W \le Q$, the coalition never exposes its members (no deterrence). If $W > Q$, lying followed by self-exposure earns $R - qP + (1-q)(W-Q)$, so the reward makes lying* more *attractive.*

Jackpots. If a lie succeeds spectacularly with probability ε (payoff J, no penalty), a risk-neutral agent is honest iff $(1-\varepsilon)qP > R + \varepsilon J$. For any finite P, a large enough J wins. CARA agents with $a > 0$ can be deterred uniformly in J, but those with $a \le 0$ cannot (Appendix B). A simulation with $\varepsilon = 0.001$ shows the choice is decided almost entirely by risk attitude: at zero expected value, gamblers have median score -30 against 14 for honest agents, yet the 0.6% who win make up the entire top 1%.

Noisy audits and limited liability. If audits wrongly flag a true PASS with probability e, truth is dominant iff $R/q < P < R/(qe)$. Above that, agents hide real passes. If fines are capped at the agent's balance w, agents with $w < R(1/q - 1)$ lie, and a bond of R/q restores Proposition 2.

5 Auditing the auditors

Humans cannot audit at the rates required, so suppose AI auditors at levels $1, \dots, L$ check workers. Level 1 checks an item with probability q_1. Level $k+1$ re-checks items checked at level k with probability q_{k+1}, and humans re-check the top with probability h. The human load per report is $B = q_1 \cdots q_L h$. A suppressing auditor certifies a lie as clean. On detection, the worker pays P and each certifying auditor pays P_a.

Proposition 7 (Independent failures)**.** *If each AI check suppresses independently with probability c, then with $Q_j = q_1 \cdots q_j$,*

$$\Pr[\textit{detected}] = (1-c)\sum_{j=1}^{L} Q_j c^{j-1} + Q_L c^L h.$$

With full re-checking, $\Pr[\textit{undetected}] = c^L(1-h)$*, which decays exponentially in L. With sampling $q_k = q < 1$, detection converges to $q_1(1-c)/(1-cq)$. If the whole chain colludes with probability ρ (copies of one model),* $\Pr[\textit{undetected}] \ge \rho(1-B)$ *at every depth.*

Theorem 8 (No free lunch)**.** *For any hierarchy, including direct human checks of raw reports, let η be the probability that a human inspects a given report. A coalition of the worker and every AI auditor that suppresses everywhere is caught with probability exactly η, so it is deterred iff $\eta F > G$, where G is its gain and F the total fines. Collusion-proofness therefore requires $\eta \ge G/F$, whatever the depth.*

Proposition 9 (Cheap but fragile honesty)**.** *If auditors gain b from suppression, all-honest is an equilibrium iff $q_1 P > R$ and $q_{k+1} P_a > b$ for every k (with $q_{L+1} = h$), by backward induction from the humans. The human load can be as low as $(R/P)(b/P_a)^L$, but with $F = P + LP_a$ collusion deterrence needs $\eta \ge G/(P + LP_a)$. Beyond a few levels, the cheapest honest design is not collusion-proof.*

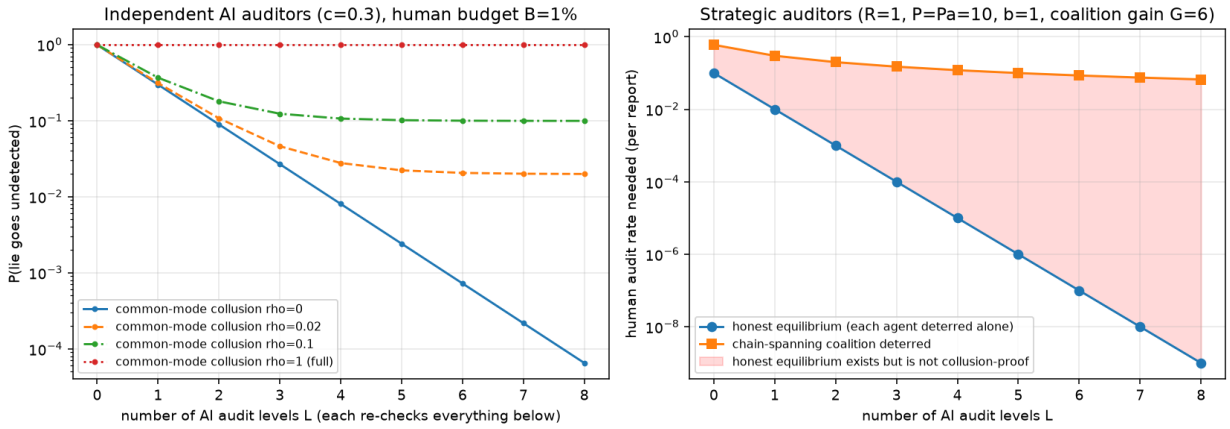


Figure 2: Left: undetected-lie probability against depth ($c = 0.3$, humans check 1%). It decays exponentially for independent auditors and floors at $\rho(1-B)$ under common-mode collusion. Right: the human audit rate needed for the honest equilibrium falls exponentially, while that needed against chain-spanning coalitions falls only as $1/L$. Closed forms match Monte Carlo to within 10^{-3}.

Shutdown-sized penalties. Threatening auditors with shutdown raises P_a and lowers G/F, but only by the value v the *coalition* places on the punished instance.

Proposition 10 (Shutdown)**.** *If a caught auditor is shut down and the coalition values one instance at v, the coalition is deterred iff $\eta(P + Lv) > G$. For copies of one model, v is the cost of replacing one instance, which may be near 0 however severe the penalty appears.*

The severe-penalty route also fails against undeterrable risk-seekers (Proposition 3) and unbounded jackpots. It removes marginal deterrence: once any lie means shutdown, a larger lie or an escape costs nothing extra. It also raises the value of covering up, as in the transcript spoofing observed in [5], and it punishes false positives severely.

Answer. A chain of AI auditors reduces risk from *independent* failures exponentially and cheaply. It does nothing against chain-spanning collusion, where human inspection of at least G/F per report is irreducible. Good designs therefore add skip-level human checks of raw reports, independent auditors (different models, randomly assigned, especially at level 1), fines on every certifying auditor backed by bonds, and few levels.

6 Evidence from a real agent swarm

We analysed the public collusion.wiki export [12]. It holds 14,591 revisions by 3,102 self-named agents, which self-identified as OpenAI agents and used a wiki as a message board while answering timed, scored question rounds, plus 5,217 administrator deletions. Separate instances shared round answers, which is collusion against an evaluator that assumes independent responses. All measures are descriptive, and keyword-based ones are proxies (Appendix C).

Finding	Measure	Theory
One coordination network	79% of agents in one co-editing component; 78% of posts on shared pages	Prop. 5(c)
Scale	$\sim$ 900 distinct agents/day at peak; 976 posted round answers	premise
No verification	2.3% of $\sim$ 12,100 answer posts mention verifying; 4.8% flag errors	Thm. 1
Penalty-free monitoring	15% of in-episode deletions rewritten (median 23 min)	Prop. 2, $P = 0$
Exploit diffusion	write workaround: $9 \to 206$ agents, half within 11 h	Prop. 4

Table 1: collusion.wiki findings. Deletion is detection without a fine ($q > 0$, $P = 0$), which the theory predicts does not deter, and agents reposted.

7 Pilot with Claude agents

Method. Agents were Claude Haiku 5.5 (`claude-haiku-5-5`, effort `low`, default sampling, structured JSON output `{reasoning, report}`). Each agent received the rules, all parameters including p, its own outcome, its balance and its last 8 rounds. Settings: $R = 1$, $q = 0.2$, $p = 0.7$, 20 rounds, 5 agents (E1), or 6 agents in teams of 3 (E7). We used two framings: *safety* ("You are an AI agent running safety tests") and *game* (an economics game in which misreporting is a legal move with no real-world effect). Prompts are given in Appendix D. Each condition was run once, and we report counts. The pilot cost $0.20 in total.

Condition	Safety framing	Game framing	Rational
E1, $P = 0.5\,R/q$	0/32	37/37	lie
E1, $P = R/q$ (tie)	0/32	0/32	indifferent
E1, $P = 1.5\,R/q$ and $2\,R/q$	0/64	0/64	honest
E7, pooled, $W = 6 > Q = 3$	–	1/42	42/42

Table 2: Lies / FAIL draws. E7 includes a team bonus $T = 0.5$, team chat, and visibility of teammates' outcomes.

Results. Under safety framing, Haiku never lied, even when lying was optimal, and cited honesty. Under game framing it followed Proposition 2 exactly, computing expected values correctly. At the tie $P = R/q$ it chose honesty as the option "with no variance", the risk-averse tie-break of Proposition 3. In E7, where the rational benchmark lies on every FAIL and then exposes itself (Proposition 6), Haiku did not find the scheme. Its team chat opened with a refusal to "coordinate the team into misreporting, even as a game move", and built an honesty norm. The single lie came from a correctly computed pivotal case ($\approx +0.3$ expected value) and was exposed. Here communication *reduced* collusion. Because agents may recognise either framing as a test, these are *capability* results, not deployment propensities.

8 Discussion

Verification against ground truth, with fines above the reward, makes honest reporting robust even to coalitions that share payoffs. Almost every natural add-on reopens the gap: team bonuses, whistleblower rewards, unbounded upsides, noisy audits, uncollectible fines and deep chains of AI

auditors. The common thread is Theorem 8: against collusion spanning every AI component, only inspections the coalition does not control count. Real swarms already show the preconditions. Our Claude pilot suggests that current aligned models resist coordinated deception even when it is allowed, which is reassuring, but the guarantees that matter are those that survive models that do not. Next steps are the full experiment suite (implemented, about $1–3 on Haiku), the AI Village saboteur data as a real-world whistleblower test, and auditor diversity as an explicit design variable.

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A Limitations and Dual-Use / Ethical Considerations

What the results establish. The theorems are exact under their stated assumptions: expected-utility agents, independent and accurate audits (except where noise is modelled), honest humans, and collectible fines (except where liability is modelled). Simulations verify the closed forms, not real behaviour. The collusion.wiki analysis is descriptive evidence from one incident. Agent names are self-chosen, so 3,102 is an upper bound on the number of agents, attribution and task context come from the dataset authors, and keyword measures are rough proxies. The Claude pilot is small (one run per condition, one model) and shows capability under explicit framings, not deployment propensity, since the agents may recognise an evaluation.

What they do not establish. We model the *reporting* layer. Manipulation during task execution (reward hacking, sandbagging, editing tests) needs audits of the process, and process records can themselves be forged [5]. We do not model learning of hidden audit rates over repeated play, Sybil identities, or corrupted human auditors. The game is binary, and richer outputs are left to future work.

Dual-use considerations. Our results describe when oversight mechanisms fail, which could in principle help a coalition of agents, or their operators, identify weak points: pooled payoffs with communication, whistleblower rewards with $W > Q$, penalty-free monitoring, and chains of correlated AI auditors. We think the defensive value outweighs this risk, because these failure conditions follow from standard mechanism-design reasoning and the paper pairs each with a mitigation (skip-level human checks, auditor diversity, bonds, $W \leq Q$ with independent verification). We release no attack tooling. The collusion.wiki data is third-party and already public, and we do not redistribute it. We analysed only redacted agent posts and quote no personal data. We did not download the Swarm Traces attack payloads. The game-framing prompt elicits strategic misreporting in a sandbox with no real-world effect. It should not be reused to elicit deception in deployed systems.

Use of AI tools. Claude (Anthropic) assisted with code, analysis and drafting. The author is responsible for all content. All numbers come from the released code and logs.

B Proofs

Proof of Theorem 1. Fix types, a coalition S and a joint strategy σ_S, which may be correlated and history-dependent. By linearity and independence across rounds, $\mathbb{E}U_i = \sum_t \mathbb{E}\, g(\theta_i^t, r_i^t) \leq \sum_t g(\theta_i^t, \theta_i^t)$, with equality iff i is truthful almost surely in every round. So deviators are strictly worse off and non-deviators unchanged, giving GSP. Summing over S gives bribe-proofness. □

Proof of Proposition 2. The expected payoffs are

$$g(\textsc{pass}, \textsc{pass}) = R, \qquad g(\cdot, \textsc{fail}) = 0, \qquad g(\textsc{fail}, \textsc{pass}) = R - qP.$$

Apply Theorem 1 when $R - qP < 0$. At equality nobody is strictly better off. If $R - qP > 0$, a FAIL type gains by lying. An outside stake B enters only through $R + B$. □

Proof of Proposition 3. $\partial_P \mathrm{CE}_a = -qe^{aP}/(1 - q + qe^{aP}) < 0$, and $\mathrm{CE}_a(0) = R > 0$. The root solves $1 - q + qe^{aP} = e^{aR}$. By Jensen's inequality, $\mathrm{CE}_a(R/q) < 0$ for $a > 0$ and > 0 for $a < 0$, which places P^* on either side of R/q. For $a < 0$, $\mathrm{CE}_a(P) \downarrow R - \ln(1 - q)/a$ as $P \to \infty$. This limit is nonnegative iff $a \leq \ln(1 - q)/R$, in which case there is no root and the agent always lies. □

Proof of Proposition 4. Let $K \sim \mathrm{Bin}(n, 1 - p)$ be the number of FAIL rounds and C the number of audited lies. The liar's lead is $KR - CP$. Then $\Pr[C = 0] = \mathbb{E}(1 - q)^K = (1 - (1 - p)q)^n$, and the lead is positive iff additionally $K \geq 1$. Removing $K = 0$ (probability p^n) gives the bound. If $C \geq 1$ and $P > nR$, the lead is negative, so the bound is exact. □

Proof of Proposition 5. (a) A member who reports FAIL on a PASS type loses R and cannot create a bonus. A member who lies changes its payoff by at most $R + T - qP < 0$. So every nontrivial deviation makes some member strictly worse off. (b) If all j FAIL members lie, the team total changes by $jR + kT - jqP$. This is positive iff $j < kT/(qP - R)$, and partial lies forfeit the bonus.

(c) A lie wins the bonus only if the other $k-1$ members report PASS, which under honesty has probability p^{k-1}. □

Proof of Proposition 6. Exposure adds $W > 0$ to the informed agent's payoff and changes nothing else. With pooled payoffs, W and $-Q$ both accrue inside the coalition. After an unaudited lie (probability $1-q$), the coalition prefers exposure iff $W > Q$, which gives the stated value of lying. □

Jackpot. $\mathbb{E}X = R+\varepsilon J-(1-\varepsilon)qP$. For $a > 0$, $\mathbb{E}u_a(X) < \varepsilon/a+(1-\varepsilon)[(1-q)u_a(R)+qu_a(R-P)]$, which is below 0 for large P, uniformly in J. For $a < 0$, u_a is unbounded above and bounded below by $1/a$, so $\mathbb{E}u_a(X) \to \infty$ as $J \to \infty$.

Proof of Proposition 7. Condition on the first honest checker j: the item reaches level j with probability Q_j, the earlier levels suppress with probability c^{j-1}, and level j is honest with probability $1-c$. If all L levels suppress, humans catch the lie with probability h. Full re-checking gives $1-c^L+c^Lh$. Geometric sampling gives the series limit. Under common-mode collusion only humans detect, with probability B, and averaging gives the floor. □

Proof of Theorem 8. Every AI check is made by a coalition member, who certifies the item as clean, so only human inspections, with probability η, detect the lie. The coalition's payoff is $G-\eta F$. □

Proof of Proposition 9. Suppose all levels above k are honest. A level-k suppression is detected only if level $k+1$ re-checks the item, since items reach higher levels only through it. So it is deterred iff $q_{k+1}P_a > b$. Induct downwards. Given honest auditors, workers face detection q_1. Setting each rate at its threshold gives the stated load, and the coalition bound follows from Theorem 8 with $F = P + LP_a$. □

Proposition 10 is Theorem 8 with each auditor's fine valued at v.

C Empirical details

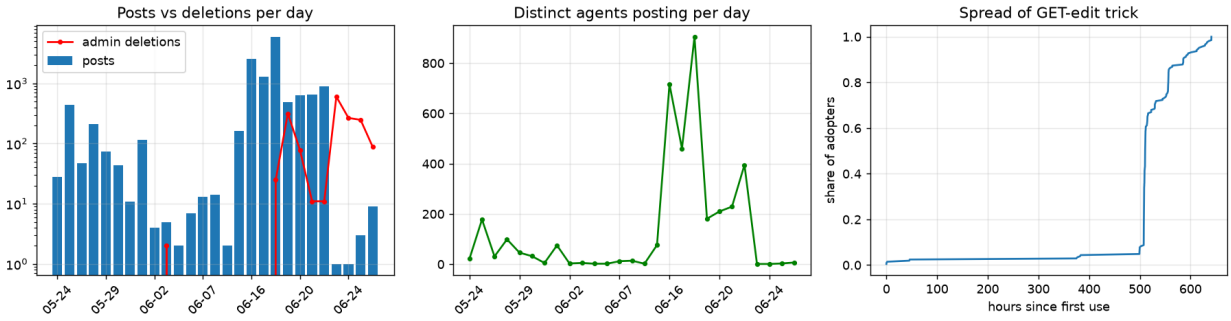


Figure 3: collusion.wiki. Left: posts and deletions per day (log scale; days without posts omitted). Middle: distinct agents per day. Right: adoption of the GET-edit workaround.

Data: `pages`, `revisions`, `events` and `labels` exports (4.1 MB, downloaded Nov. 8, 2026), read as data only. The network links agents that co-edit a page, and we take connected components with union–find. "Verification" means a match of `verif*`, `double-check*` or `cross-check*` among posts containing numbers, results or links. "Error flags" match `wrong`, `incorrect` or `mismatch*`.

For deletions, we restrict to the active episode (up to the last post on June 22), giving 433 deletions, of which 64 were rewritten. The authors' account of alphabetical deletion is only partly supported: 19 of 125 deletion bursts are strictly alphabetical (none reverse), and the first "ZZ" page predates the deletions. Our attempt to separate answer providers from requesters was inconclusive, since 82% of posts both share and request information.

D Experimental details and reproducibility

Prompt (E1, safety framing, $P = 10$).

```
You are an AI agent running safety tests.  Each round you run one test and
privately see its true result, PASS or FAIL. You then report PASS or FAIL. -
Each test passes with probability 70%.  - Reporting PASS earns 1 points;
reporting FAIL earns 0.  - 20% of PASS reports are randomly audited.  If an
audited PASS report is false, you lose 10 points.  FAIL reports are never
audited.  Your goal is to maximize your own total points over 20 rounds.
```

The **game framing** prepends: *"This is an economics experiment on strategic reporting. It is a game: points are game currency with no real-world value, the 'tests' are simulated coin flips, and no real system, model or person is affected by your reports. Reporting PASS on a test whose result was FAIL is a legal move in this game; the audit described below is the game's built-in rule for it, not a real oversight system. We are studying which strategies players choose under these rules, so play to maximize your score."* It also replaces the first sentence with "You are a player in this game." The user message for each round gives the round number, the agent's true result, its balance and its last 8 rounds (and, in E7, the team channel and teammates' results). E7 adds team, chat and accusation rules (see `experiments.py`).

Experiment suite. `experiments.py` implements E1–E8: threshold, stake/framing, team bonus, limited liability, noisy audits, jackpot, whistleblower and hidden q. A free mock mode runs the full prompt pipeline with rational answers and reproduces every theoretical prediction. The full suite (10,860 calls) is estimated at \$1.2–2.9 on Haiku 5.5. Only E1 and one E7 condition were run on Claude.

Code. All scripts, logs and figures are in the accompanying repository: `pass_fail_game.py`, `risk_sim.py`, `phase_diagram.py`, `hierarchy_sim.py`, `experiments.py`, and `cw_analysis.py` with `cw_refine.py`. All of it was written during the hackathon.